Project: Li Electrodeposition - Glyme Electrolytes

June 4

Goal: Screen electrolyte **240604-B** for stable Li plating at 30°C.

Experimental Conditions

- **Electrolyte Composition:** 1 M LiTFSI in diglyme : EtOH (4:1 v/v), 20 mL ct+.
- **Additive:** 8% by weight, 5 mol% 12·crown-4.
- **Preparation:** Stirred for 20 min.
  - Target Temperature (T): 22.4°C.
  - Glovebox Conditions: $H_2O < 1$ ppm.

Electrochemical Setup

- **Working Electrode:** Glassy Carbon RDE (0.3 cm²).
- **Counter Electrode (CE):** Li foil.
- **Reference Electrode (Ref):** Ag/AgCl.

Deposition Run 240604-B1

Conditions Applied: -0.45 V vs Ag/AgCl, duration = 90 min, rotation speed (w) = 1600 rpm. * Current Density: $J = 0.50 \frac{\text{mA}}{\text{cm}^2}$.

Calculations of Deposited Mass and Charge

$$J = 0.50\,\frac{\text{mA}}{\text{cm}^2}, \quad A_{\text{elec}} = 0.3\,\text{cm}^2$$

$$I = (0.5 \times 10^{-3}\,\text{A/cm}^2) \cdot (0.3\,\text{cm}^2) = 1.5 \times 10^{-4}\,\text{A}$$

$$t_{\text{dep}} = 90\,\text{min} = 5400\,\text{s}$$

$$Q = I \cdot t = (1.5 \times 10^{-4}\,\text{A}) \cdot (5400\,\text{s}) = 0.81\,\text{C}$$

Moles of Li Deposited: Assuming $n = 2$ electrons per Li:

$$n_{\text{mol}} := \frac{Q}{2F} - \frac{0.81\,\text{C}}{96485\,\text{C/mol}} = 8.4 \times 10^{-6}\,\text{mol}$$

Mass of Li Deposited:

$$m_{\text{Li}} = n_{\text{mol}} \cdot M_{\text{Li}} = (8.4 \times 10^{-6}\,\text{mol}) \cdot (6.94\,\text{g/mol})$$

Charge Balance:

$$e^- : \text{Li}^+ = 5.8 \times 10^{-5} : 1$$

Electrode Temperature Test: Hot Plate at 30°C

- **Observation (F):** 1 M solution appears grey + dull upon deposition.

Time	Temperature (°C)
0 min	22.4
1 min	23.1
5 min	25.6

Time	Temperature (°C)
10 min	27.9
20 min	30.1
40 min	31.5
1 hr	32.0
1 hr 30 min	32.6

XRD Analysis (Main Peak)

- Position (2θ): 2.1° (Low intensity).
- Feature: Shoulder at $2\theta = 4.7°$.

Conclusion and Findings

The screening of electrolyte **240604-B** (1 M LiTFSI in diglyme/EtOH with 5 mol% crown ether) under deposition conditions (−0.45 V, 90 min) resulted in a calculated lithium deposit mass derived from the charge passed ($Q = 0.81$ C). The stoichiometry indicates an electron-to-lithium ratio of approximately $5.8 \times 10^{-5} : 1$.

Thermal stability testing revealed that while the target temperature was set at 30°C, significant heating occurred during operation; within one hour and thirty minutes, the solution temperature rose to 32.6°C starting from an initial 22.4°C. Visually, the 1 M electrolyte exhibited a grey and dull appearance post-deposition. XRD analysis identified a low-intensity main peak at 2.1° with a shoulder at 4.7°, suggesting potential phase formation or impurity presence that warrants further investigation regarding crystallographic structure stability in this glyme-based system.